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Customer Churn Analysis for the American Telecommunications Company

COMM3501 INDIVIDUAL ASSIGNMENT REPORT

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1. Introduction and Business Context

The telecom industry is highly saturated and competitive, driven by rapid technological advancement and shifting customer preferences. This led to consistently high churn rates across the telecom industry, both in America and globally. Losing a customer can result in substantial financial loss and brand damage. According to research, acquiring a new customer costs 6–7 times more than retaining an existing one, while reducing churn by just 1% can boost profits by 5% (Waweru 2025), illustrating the importance of customer retention.

Therefore, for the American wireless telecom company providing cellular telephone service such as our client, this report will explore the key factors driving customer churn to support the development of a proactive retention program. This is achieved through predictive modelling, decision optimisation for retention strategy, and associated risk assessment. By targeting the right customers and focusing resources on high-risk segments, the client can improve retention rates, build a more loyal customer base, and thus increase customer lifetime value.

2. Data Cleaning and Manipulation

To determine and predict the key factors influencing customer churn, a training set of 30,000 observations and 44 variables and evaluation set of 10,000 observations with 43 variables were used. Both datasets were confirmed free of duplicates through full-row and unique-customer-ID checks. All categorical variables were transformed to factors, with binary variables encoded as 0/1 for ease of modelling.

Investigating data quality, it's found that both datasets have missing values (Appendix 1.1). Two distinct systematic patterns are identified, indicating a missing-at-random (MAR) situation likely caused by data input errors. The same 91 rows were missing all five usage and revenue variables. Dropping this pattern would risk biasing the model, as these observations display a 41.8% churn rate, notably higher than the baseline. Therefore, they're flagged to include in modelling. Another pattern involved 520 observations with missing age values, likely due to an unmodified default setting. Additional inconsistencies were identified and resolved, including invalid zero-filled age values (recoded to NA), active-subscription accounts with $0 recorded revenue (retained and flagged), and minor missing values across several service and billing fields encoded to NA for imputation. A household composition flag (single- or two-adult) was also derived from the age fields prior to resolving the age-related pattern (Appendix 1.2 and 1.3).

For modelling, this training dataset was partitioned using the 70:30 split. Missing categorical and numerical variables were imputed using mode and median respectively, with median chosen given the heavy right-skew of numeric distributions inspected through histograms (Appendix 1.4). All imputations used the training set's central tendency measures only to prevent data leakage (Appendix 1.5). As the skewed distributions could also bias predictions and violate modelling assumptions, all numerical continuous variables were log-transformed. To reduce multicollinearity, *DroppedBlockedCall* and *MadeCallToRetentionTeam* were removed due to high correlation and overlapping information with other usage- and retention-related variables (Appendix 1.6)

To translate raw data into meaningful patterns, several features were engineered around tenure, usage, and demographics (details in Appendix 1.7). Service reliability is a key driver of customer satisfaction

(ACMA 2020), therefore *TotalCallIssue* and *CallIssueRate* were constructed to capture this pattern. *ServiceArea* was also decomposed into three distinct identifiers (market, locality and telephone area code), though only market was analysed further, given the high cardinality of the other two.

The evaluation dataset was cleaned and transformed using the same method, with imputation values, binning thresholds, and derived features computed exclusively from the training data, ensuring no information leakage between the two sets. Unseen data and derived variables defaulted to NA in this evaluation set were handled specifically to preserve consistency with the training-derived methods (Appendix 1.8).

3. Descriptive Analysis

Before modelling, it is first essential to understand the characteristics of customers and how they relate to churn behaviours.

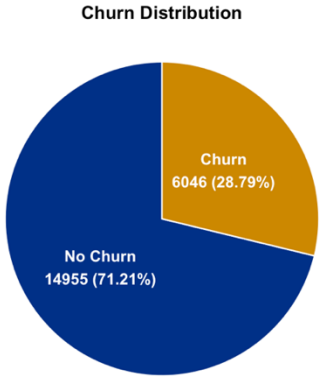


Figure 1: Churn Distribution

The company has an average churn rate of 28.79%, surpassing the industry benchmark of 21.5% (McGee 2026). This indicates that the company is underperforming relative to competitors, with a gap of 7.29 percentage points to close.

Among customer demographics, service area market stands out as the most notable indicator. Of 55 market types, the top two, Hawaii (HWI) and Salt Lake City, Utah (SLC), contribute the most to churn with a churn rate exceeding 40%. The top-churn markets show a rate more than double that of the bottom 10 markets, suggesting service area may be strongly associated with churn potentially due to service reliability or local competition. Regarding tenure, churn is the highest among customers who have stayed between one and two years, gradually decreasing with longer tenure but remaining above the industry average. Only new customers (under one year) show an acceptable churn rate, suggesting that the company's acquisition and onboarding tactics are effective. However, service and support quality may decline in the customer's perception the longer they stay with the company.

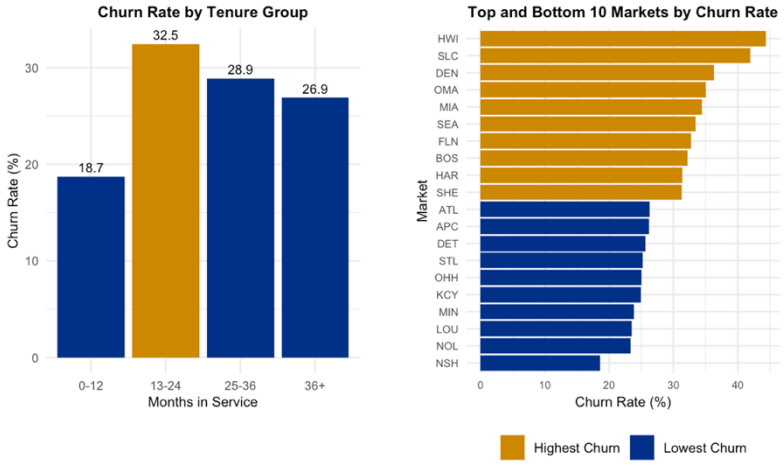


Figure 2 &3: Churn Rate by Tenure Group and Service Area Market

In terms of usage patterns, churn rate is relatively flat across forwarding and roaming calls, while other call-related variables show a clear downward trend with churn, indicating call issues and service quality might be key determinator of churn. Examining the client's existing retention efforts, a non-monotonic trend is evident: customers who received one or two retention offers are still churning at high rates. This raises the question of whether retention offers have been targeting the right customers, and whether these offers align with what customers expect from the company.

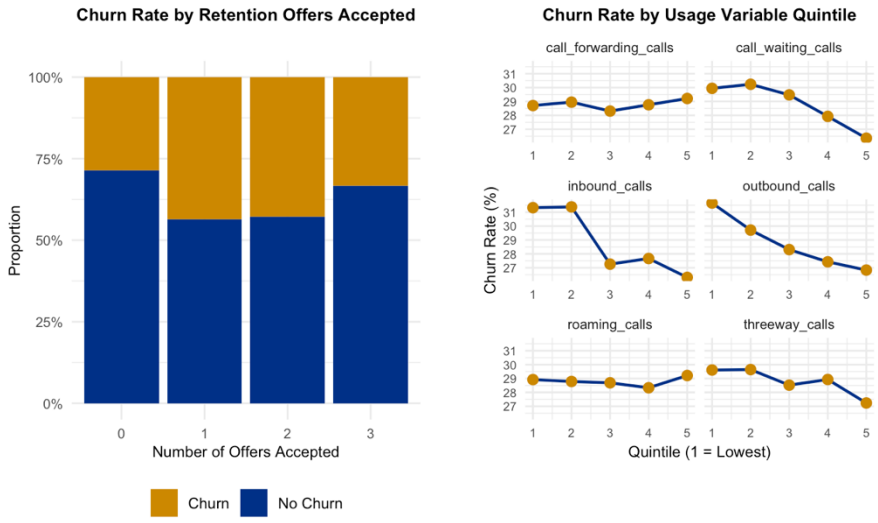


Figure 4 &5: Churn Rate by Retention Offers Accepted and Usage Level across Services

Together with predictive modelling, this analysis provides insight into the company's current performance in communication and customer service delivery.

4. Predictive Modelling and Analysis

4.1. Predictive Modelling

For this case, the priority is minimising false negatives, or churned customers incorrectly predicted as retained. The core objective is to identify as many churned customers as possible, even at the cost of including more false positives. With the client's cost structure, offering retention tactics to a falsely predicted non-churn customer results in an opportunity cost that is considerably lower than the cost of losing a customer. Model selection therefore prioritises AUC, recall/sensitivity, F1, and Kappa. Since churn is imbalanced, AUC is assessed using both ROC and PR curves to give a more complete picture of each model's predictive power.

A simple logistic regression (LR) model was first fitted using all predictors, excluding high-cardinality and raw-count variables (Appendix 2.1), to establish a baseline and assess variable importance. At a 5% significance level, only 32 of 45 predictors were statistically significant (Appendix 2.2). Then, another LR was fitted using only significant variables ("Log Selection"). Overall, LR "All" performed marginally better than LR "Selection," indicating that non-significant variables still contribute some predictive value to the model (details in Appendix 2.3). LR "All" achieved an AUC of 0.612 (ROC) and 0.371 (PR), with a Kappa of 0.0295, slightly better than random guessing. At the default 50% cutoff, recall was just 3.5%, low but expected given the level of class imbalance.

Moving on to random forest (RF), four variants were constructed: all variables ("All"), LR-significant variables ("Log Selection"), the top 14 variables by Gini index ("Gini"), and variables confirmed by the Boruta method ("Boruta"). The Gini-based variant was tested on the assumption that excluding variables with relatively low impurity reduction (below 50) might yield a more efficient model. Whereas, Boruta was used to confirm all genuinely relevant variables through its handling of complex interactions, avoiding omitting meaningful predictors as in the “Log Selection” approach (detailed variables in Appendix 2.4).

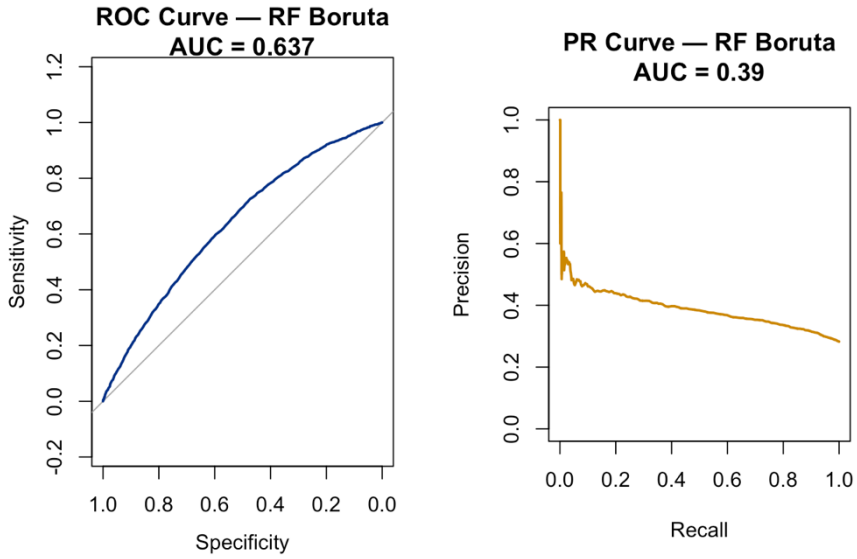


Figure 6: ROC and PR Curves of the best Random Forest model (RF “Boruta”)

Overall, RF "Boruta" performed best among the four variants, achieving an AUC of 0.637 (ROC) and 0.39 (PR). Not only predictive power improved, but Kappa and Recall were also slightly higher for RF at the default threshold (0.038 and 4.02% respectively). As RF performance is largely threshold-dependent, the default cutoff might understate the model's usable predictive power, which could be improved with threshold tuning (details in Appendix 2.5)

To test whether a more complex gradient boosting approach improved performance, an XGBoost model was fitted using all variables. AUC improved only marginally to 0.639 (ROC), while PR-AUC plateaued at 0.39, similar to RF “Boruta” (Appendix 2.6). Given this performance, the added complexity and reduced interpretability of XGBoost, model selection was narrowed to LR "All" and RF "Boruta". As the dataset is imbalanced, threshold tuning is essential to improve the model performance. Besides the default threshold, Youden’s J and F1 were used as the tuning criteria. Youden's J identifies the cut-point maximizing Sensitivity + Specificity − 1 under equal weighting of both error types (Ruopp et al., 2008), while F1 balances precision and recall for the churn class specifically.

Both statistics improve performance considerably, with F1-optimised RF performing best on recall, the primary metric. At this threshold of 0.26, close to the churn prevalence rate, recall reached 77.4%, a strong result given the model's overall predictive power. The resulting confusion matrix shows false negatives minimised to 562, at the cost of the highest false positive count observed (Appendix 2.7)

Model	Threshold	Sensitivity	Precision	F1	Specificity	Accuracy
Logistic Regression	Default (0.5)	0.035	0.503	0.065	0.987	0.718
Logistic Regression	Youden's J (0.286)	0.576	0.354	0.439	0.587	0.584
Logistic Regression	F1 (0.24)	0.771	0.327	0.459	0.376	0.487
Random Forest	Default (0.5)	0.040	0.549	0.075	0.987	0.720
Random Forest	Youden's J (0.283)	0.697	0.356	0.472	0.505	0.559
Random Forest	**F1 (0.26)**	**0.774**	**0.343**	**0.475**	**0.417**	**0.518**

Figure 7: Performance Metrics Comparison after Threshold Tuning

4.2. Selected Model Interpretation

Considering both post-tuning performance and overall predictive power, RF "Boruta" is the best model available (full comparison in Appendix 2.8). The decile-wise chart shows the model performs meaningfully above random guessing only within the first five deciles, reaching a maximum lift of 1.56 times. This is a moderate result, consistent with the predictive power shown in the ROC and PR curves.

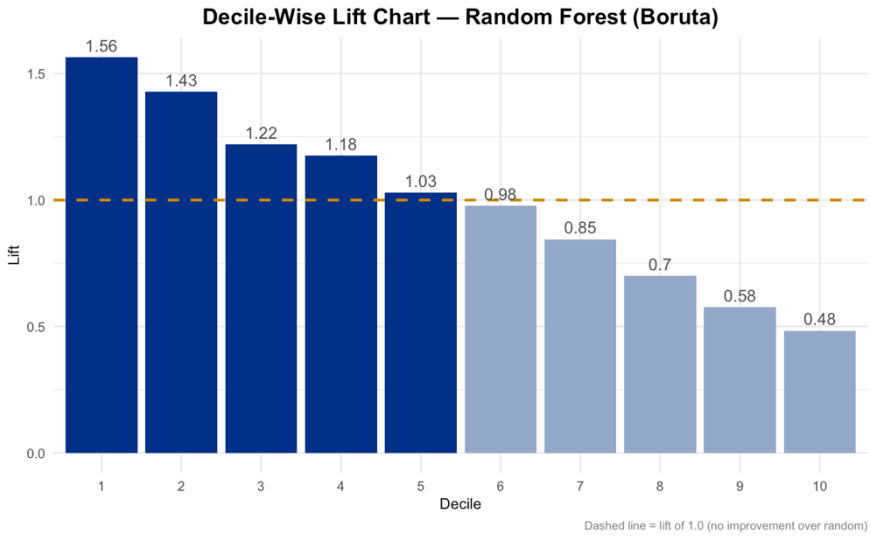


Figure 8: Decile-Wise Lift Chart of RF "Boruta" Model

The full list of variables and their importance scores is provided in Appendix 2.9. In summary, customer churn appears to be significantly influenced by:

1. **Equipment and tenure:** *CurrentEquipmentDays, MonthsInService, Handsets, EquipmentAgeRelativeToTenure*
2. **Usage and call activity:** *MonthlyMinutes, OverageMinutes, OutboundCalls, InboundCalls, RoamingCalls, CallWaitingCalls*
3. **Service issues:** *CallIssueRate, UnansweredCalls, BlockedCalls, DroppedCalls, CustomerCareCalls*
4. **Billing and product pricing:** *MonthlyRevenue, TotalRecurringCharge, RevenuePerMinute, OverageRate*
5. **Customer's seniority:** *AgeHH1*

To examine these predictors in more depth, three break-down plots were constructed for customer segments defined by median churn probability: typical churner, typical non-churner, and borderline. Variable contributions were computed using the break-down decomposition method, which sequentially

attributes predicted probability to individual predictors in order of importance. As the model uses log-transformed values, which are beneficial for modelling but not directly interpretable, all numeric variables were converted back to their original scale. Using the top 10 predictors, the three profiles are summarised below (details in Appendix 2.10):

	Typical Churner (p = 0.462)	**Borderline (p = 0.26)**	**Typical Non-churner (p = 0.13)**
Top Risk Drivers	• Longer equipment days • High overage rate • Low revenue and usage level • Low tenure (belongs to the highest churn group)	• Moderate equipment days (less than median – decrease risks) • High revenue per minute, but low revenue due to very low usage level • High call issue level	• Moderate equipment days (relatively similar to Borderline – decrease risk) • Low tenure (belongs to the lowest churn group – decrease risk) • Moderate revenue and usage level
Demographics	• Credit Rating: 6	• Credit Rating: 6	• Credit Rating: 5 • Low Income (Group 1)

Figure 9: Customer Segments based on Churn Probability

Equipment age appears across all three profiles, confirming its overall significance in identifying churn. However, it does not clearly separate the borderline and typical non-churner profiles, suggesting that for operational targeting of at-risk customers, this variable alone may not be sufficient. A similar pattern appears with credit rating in churner and borderline profiles, indicating credit rating is a weak standalone discriminator once usage and billing behaviours are accounted for.

The churner profile is driven by a plan-usage mismatch: high overage rate suggests their current bundle no longer fits their usage pattern. This customer also falls within the 13–24-month tenure window, the highest-risk group identified in Section 3, confirming its importance in identifying churners. For borderline group, this profile suggests a poor value-for-money issue where even at low usage level and high prices, customers still experience high call issue rate. Both profiles are worthwhile targets for pricing and plan-fit intervention. By contrast, the non-churner presents a stable profile with moderate usage and revenue despite a relatively short tenure, suggesting plan fit may be a protective factor worth investigating further for retention tactics.

As personas represent illustrative individual customers rather than segment averages, specific driver values may vary slightly across model refits, though broader patterns remain consistent. Together, these segment-level findings support differentiated retention strategies for each customer type, discussed further in Section 7.

5. Decision Optimisation

For each customer, the client's profitability analysis implies a choice between two actions: offering no retention incentive, which yields expected profit of $0.15R_i$ if the customer stays and \$0 if they churn, or offering a 5% bill discount, which reduces churn probability by 90% but caps profit margin at $0.10R_i$ if retained. The opportunity cost of offering the discount to a customer who would have stayed regardless is $0.05R_i$ per year. Using each customer's predicted churn probability (p_i) from the selected Random Forest model and their predicted annual revenue ($R_i = 12 \times$ *MonthlyRevenue*), expected profit was calculated under both scenarios of no offer and offer.

Solving for the break-even churn probability where these two expressions are equal yields a threshold of p = 0.3571. This threshold was also confirmed empirically by sweeping candidate thresholds from 0 to 1 and computing total portfolio profit across the evaluation dataset at each point. Above this threshold, offering the discount is expected to generate additional profit; whereas below this, it results in a net loss.

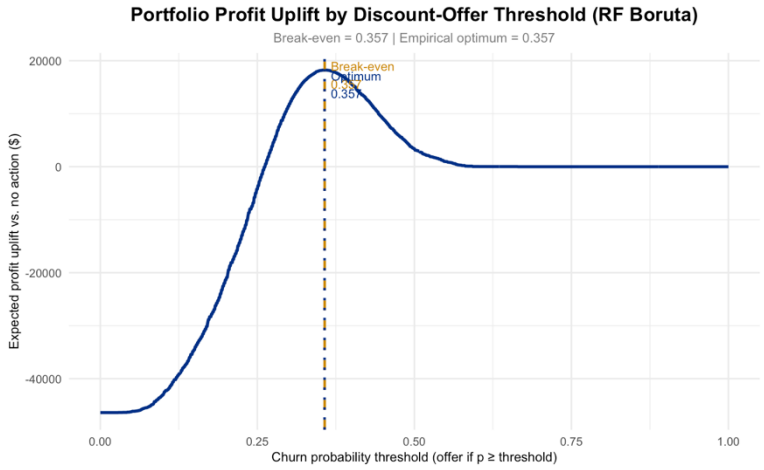


Figure 10: Sensitivity Chart of Profit Uplift relative to Threshold

At this threshold, 3,037 customers are suggested to contact, which is beyond the client's operational capacity of 3,000. The sensitivity analysis shows that below this threshold, discounts are wasted on low-risk customers, so profit rises steeply as the threshold approaches its optimum. Beyond the optimal threshold, profit declines as genuinely profitable customers who would stay if offered the discount are excluded, falling to zero once the threshold exceeds 62.4%, the highest churn probability recorded in the dataset.

Offering the discount to these 3,037 customers is expected to generate a profit uplift of $18,235.88 relative to a no-action baseline. Capping the discount to the full operational capacity of 3,000 customers would reduce this profit by only $1.72, therefore we recommend contacting only 3,000 customers for operational efficiency (Appendix 3).

6. Stochastic Modelling and Analysis

To assess the risks associated with this retention program, a Monte Carlo simulation was run to quantify potential outcomes. Monte Carlo simulation was chosen over other stochastic methods, such as parametric bootstrap, because it directly quantifies potential outcomes given known uncertainties, whereas bootstrapping measures the variability of the estimates themselves (Beasley & Rodgers 2012). In this case, as the simulation's inputs (annual revenue and churn probability under the discount offer) are treated as fixed and known, Monte Carlo simulation was the more efficient choice. To quantify the level of uncertainty in the simulation results, a bootstrap resampling method was applied to the simulated outputs to compute standard errors (SE).

Under this simulation, each customer's churn outcome is modelled as an independent Bernoulli trial with probability *p_effective*, consistent with the binary nature of the churn outcome. According to Section 5, all 3,000 highest-risk customers were offered the retention discount per the optimal threshold identified. Simulation size was determined using a convergence analysis of total churns and total profit

across run counts (Appendix 4.1). Total churns converged and stabilised after approximately 250 runs, but total profit continued to change until around 10,000 runs. Therefore, 10000 runs were used for stable and conservative estimates.

The simulation indicates a 95th-percentile ("1-in-20 worst case") total of 144 churns (SE = 0.48), and a 5th-percentile total profit of $ 200,790.3 (SE = 21.97). Neither figure deviates substantially from the corresponding mean estimate, suggesting that risk is well contained and averaged out across 3,000 customers. The profit distribution appears closer to normal than the churn distribution, though neither is heavily skewed (Appendix 4.2). These results validate the discount optimization strategy, since even under realistic outcome randomness, the strategy doesn't produce a catastrophic downside. However, Monte Carlo assume independence among customers, which isn't accurate in practice as churn events can be correlated such as a regional outage or a competitor promotion. Consequently, this result might underestimate the heavy tail risk.

7. Recommendations

Using the Random Forest model with Boruta-selected variables, the top 3,000 highest-risk customers were segmented for targeted retention tactics. For the current discount-based retention program, we recommend offering a 5% discount to 3,000 customers (capped by operational capacity) exceeding the threshold of 0.3571, yielding an expected profit uplift of $ 18,234.16. Accounting for tail risk, this retention lever remains well within the profitable range, with minimal standard error.

However, based on the descriptive and predictive modelling findings, we recommend complementing this with the following non-discount tactics:

1. **Proactive engagement and outreach** for customers approaching or past their first year of service, focusing on customers whose equipment tenure is disproportionately long relative to their service tenure.
2. **Offer an overage bundle or exclusive plan-tier upgrade** for customers with high overage rates relative to their plan before the cumulative excess charges affect their retention.
3. **Review plan pricing for low-usage customers**, offering a smaller, lower-cost bundle to target borderline customers who may be paying excessively for a plan that no longer matches their actual needs.
4. **Offer device upgrade discounts** for customers approaching a moderate equipment age (around the "Borderline" segment's median), encourage earlier device upgrade before customers reach the equipment age range associated with higher churn risk.

These recommendations are based on the accuracy of the predictions in Section 4, and consequently on the goodness of fit of the selected predictive model and the churn-probability-based customer segmentation. Given the RF model's performance and the information ceiling observed when testing more complex methods, there may be additional churn-relevant factors not captured in the current dataset. Incorporating customer-reported data (satisfaction surveys) or service-quality indicators (issue frequency, outage, or network congestion logs) would allow the model to learn directly from customer satisfaction and service experience, potentially improving predictive power.

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AI Usage Acknowledgement

I, Minh Khanh Tang, acknowledge the use of Generative AI (Claude, Anthropic) for code refactoring and debugging during the modelling pipeline, and for copy-editing and style polishing of report draft. All AI-assisted outputs were critically reviewed, adjusted, and cross-checked by the author before inclusion in the final report.

9. Appendices

Appendix 1 – Data Manipulation

Appendix 1.1: Table of Missing Data

Predictor Variable	No. of Missing Rows in Training Set	No. of Missing Rows in Evaluation Set
monthly_revenue	91	27
monthly_minutes	91	27
total_recurring_charge	91	27
overage_minutes	91	27
roaming_calls	91	27
service_area	13	7
handsets	1	NA
current_equipment_days	1	NA
age_hh1	520	183
age_hh2	520	183

Appendix 1.2: Inconsistency Check and Resolution

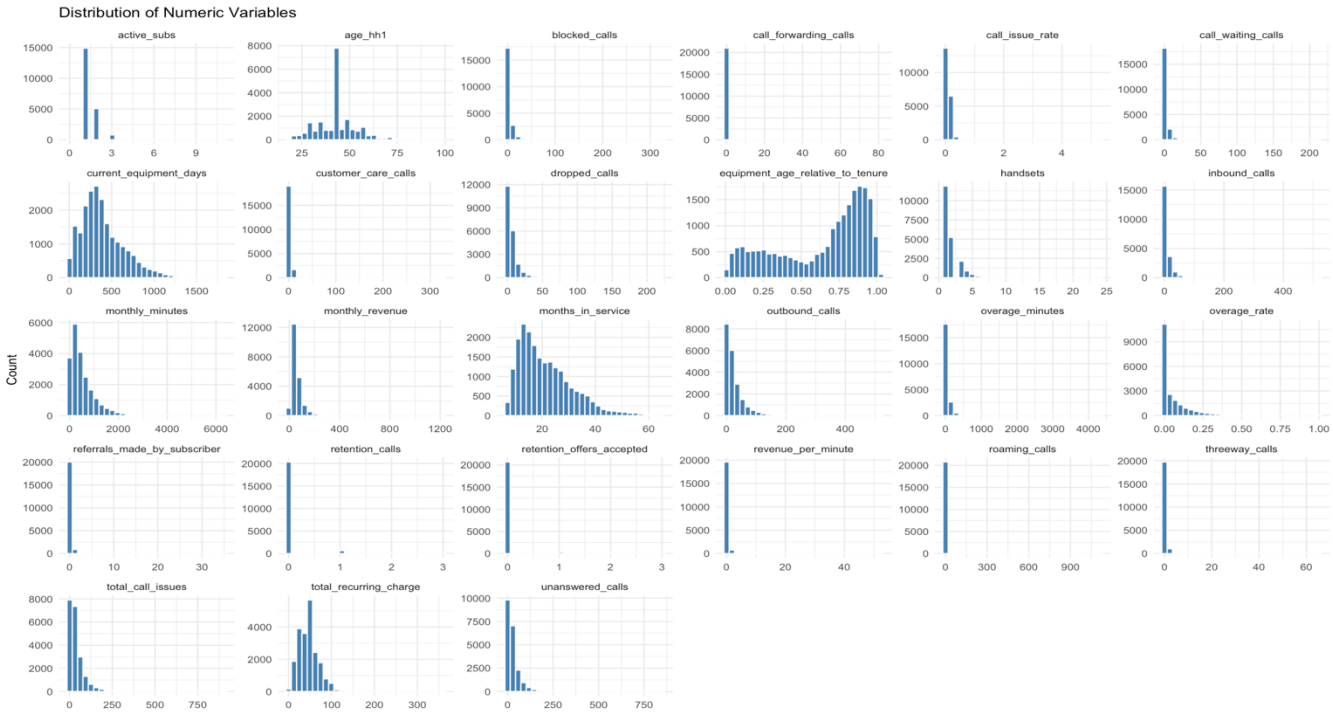
Inconsistencies Identified	Rows Affected		Resolution
	Training Set	Evaluation Set	
Zero value age fields	15348	5116	Recoded to NA prior to imputation Flagged via *age_unknown* where both AgeHH1 and AgeHH2 were 0/missing.
Negative *TotalRecurringCharge*	3	3	Recoded to NA and imputed
Negative *MonthlyRevenue*	1	1	Recoded to NA and imputed
Active subscriptions with zero *MonthlyRevenue*	4	1	Retained (not treated as missing/invalid) Flagged via *zero_revenue_active* for potential inclusion in modelling.

Appendix 1.3: Two-adult Household Rule and Flag

As two-adult households may adopt different purchasing patterns than single-adult households, a household composition flag was derived from the 0/1 pattern across AgeHH1 and AgeHH2, prior to imputation removing this signal. As no cases existed where AgeHH1 was 0 while AgeHH2 was populated, AgeHH1 was retained as the primary age variable and AgeHH2 was dropped. Below is a table illustrating three representative cases based on this flag creation:

Household Type	*household_has_two_adults*	*age_unknown*	Rows Affected
Single-adult	0	0	7139
Two-adult	1	0	14132
Missing Data	0	1	8729

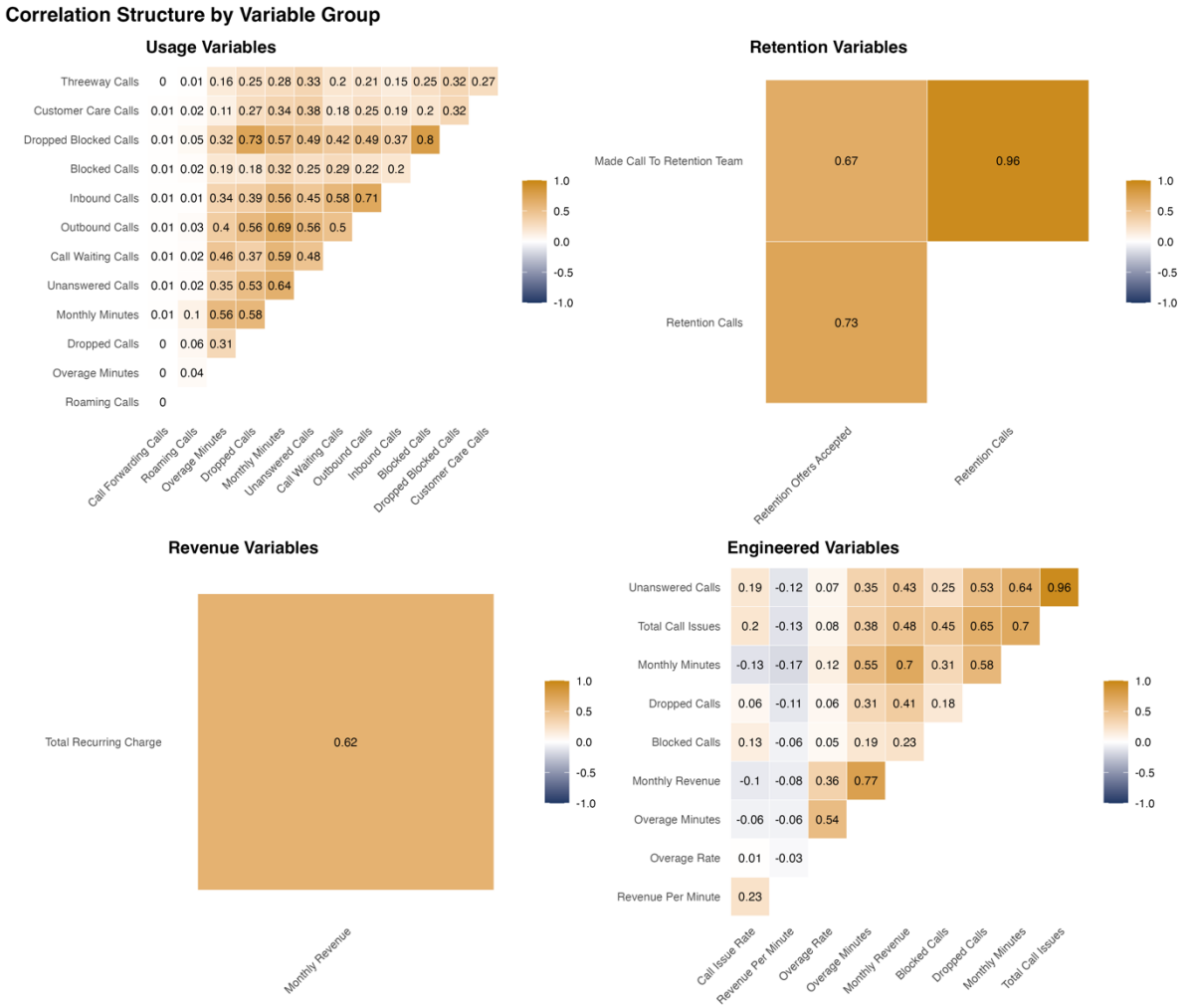
Appendix 1.4: Numerical Variable Histograms



Appendix 1.5: Data Imputation Values

Predictor Variable	Central Tendency Measure	Imputation Values (Based on Train Data)
monthly_revenue	Median	48.4
monthly_minutes		366
total_recurring_charge		45
overage_minutes		3
roaming_calls		0
handsets		1
current_equipment_days		343
age_hh1		44
service_area	Mode	NYCBRO917

Appendix 1.6: Correlation Matrix



Appendix 1.7: Table of Detailed Feature Engineering and Flags

New Feature	Engineering Method	Description
TotalCallIssues	Sum: DroppedCalls + BlockedCalls + UnansweredCalls	Aggregates individual call-failure types into a single measure of service disruption experienced by the customer
CallIssueRate	TotalCallIssues / MonthlyMinutes	Normalises call issues by usage volume, so complaint burden is comparable across low- and high-usage customers
RevenuePerMinute	MonthlyRevenue / MonthlyMinutes	Measures billing efficiency/value — how much revenue is generated per minute of usage
OverageRate	OverageMinutes / MonthlyMinutes	Proportion of usage exceeding plan allowance, relative to total usage
EquipmentAgeRelativeToTenure	CurrentEquipmentDays / (MonthsInService × 30)	Ratio indicating whether a customer is holding disproportionately old equipment relative to their tenure, i.e. whether they have upgraded within their service period

TenureGroup	Binning: MonthsInService binned into 0–12, 13–24, 25–36, 36+	Groups continuous tenure into categories for exploratory analysis of non-linear churn patterns across customer lifecycle stages
UsageLevel	Binning: MonthlyMinutes binned into Low/Medium/High	Groups continuous usage into categories to examine non-monotonic churn patterns that could inform modelling choices
SAMarket	String extraction: first 3 characters of ServiceArea	Identifies the broad telecom market (55 levels); used in modelling
SALocality	String extraction: middle 3 characters of ServiceArea	Identifies the local service area; EDA only, excluded from modelling due to high cardinality
SAAreaCode	String extraction: last 3 characters of ServiceArea	Identifies the NPA telephone area code; EDA only, excluded from modelling due to high cardinality
Household_has_two_adults	Flag: 1 if both AgeHH1 and AgeHH2 are valid non-zero values, else 0	Captures household composition (two-adult vs. single-adult)
Age_unknown	Flag: 1 if AgeHH1 and AgeHH2 are both 0/missing, else 0	Flags rows where household age data is entirely unavailable
Zero_revenue_active	Flag: 1 if ActiveSubs > 0 and MonthlyRevenue = 0, else 0	Flags active-subscription accounts with no recorded revenue
Usage_data_imputed	Flag: 1 if all five usage/revenue variables are missing simultaneously, else 0	Flags the systematic batch-missingness pattern in usage and revenue

Appendix 1.8: Evaluation Set Cleaning and Resolution

For evaluation set, 26 records fell into service area markets not observed in training data and so could not be categorised by churn rate. Therefore, *SAMarketGroup* is defaulted to “Medium” as a placeholder, on the assumption that an unknown market is closer to average risk than to the extremes. 135 rows of derived variables (*CallIssueRate, RevenuePerMinute and OverageRate*) also defaulted to NA where *MonthlyMinutes* was zero, thus could not be calculated and were imputed with 0.

Appendix 2 – Predictive Modelling

Appendix 2.1: List of All Predictors used across all models

1. "children_in_hh"
2. "handset_refurbished"
3. "handset_web_capable"
4. "truck_owner"
5. "owns_computer"
6. "has_credit_card"
7. "referrals_made_by_subscriber"
8. "income_group"
9. "owns_motorcycle"
10. "credit_rating"
11. "prizm_code"
12. "occupation"
13. "marital_status"
14. "age_unknown"
15. "log_monthly_revenue"
16. "log_monthly_minutes"
17. "log_total_recurring_charge"
18. "log_overage_minutes"
19. "log_roaming_calls"
20. "log_dropped_calls"
21. "log_blocked_calls"
22. "log_unanswered_calls"
23. "log_customer_care_calls"
24. "rv_owner"
25. "buys_via_mail_order"
26. "responds_to_mail_offers"
27. "opt_out_mailings"
28. "log_threeway_calls"
29. "log_outbound_calls"
30. "log_inbound_calls"
31. "log_call_forwarding_calls"
32. "log_call_waiting_calls"
33. "log_months_in_service"
34. "log_active_subs"
35. "log_handsets"
36. "log_current_equipment_days"
37. "log_age_hh1"
38. "log_retention_calls"
39. "log_retention_offers_accepted"
40. "log_referrals_made_by_subscriber"
41. "log_call_issue_rate"
42. "log_revenue_per_minute"
43. "log_overage_rate"
44. "log_equipment_age_relative_to_tenure"
45. "sa_market_group"

Appendix 2.2: Table of Logistic Regression Significant Predictors

Predictor Variable	Estimate	Std. Error	Z value	P value
sa_market_groupHigh	0.3508421	0.0449035	7.813237	0.0000000
log_retention_calls	1.0745416	0.1623662	6.618015	0.0000000
log_overage_minutes	0.0758029	0.0156185	4.853418	0.0000012
log_current_equipment_days	0.3416718	0.0771590	4.428150	0.0000095
log_age_hh1	-0.3202386	0.0735607	-4.353390	0.0000134
log_monthly_revenue	0.3109509	0.0719905	4.319330	0.0000157
handset_refurbished1	0.2056820	0.0484585	4.244503	0.0000219
children_in_hh1	0.1789326	0.0432905	4.133300	0.0000358
(Intercept)	-1.7260825	0.4197599	-4.112071	0.0000392
credit_rating.C	-0.2500278	0.0636130	-3.930452	0.0000848
log_call_issue_rate	1.1147670	0.2845147	3.918135	0.0000892
sa_market_groupMedium	0.1497381	0.0401871	3.726023	0.0001945
handset_web_capable1	-0.2043827	0.0553080	-3.695358	0.0002196
log_dropped_calls	0.0833423	0.0259716	3.208982	0.0013321
log_total_recurring_charge	-0.1821623	0.0577883	-3.152234	0.0016203
log_unanswered_calls	-0.0713657	0.0243008	-2.936765	0.0033166
log_active_subs	0.2251807	0.0768205	2.931260	0.0033759

log_handsets	0.2076561	0.0773534	2.684512	0.0072636
log_monthly_minutes	-0.1170338	0.0440476	-2.656984	0.0078843
credit_rating.Q	0.1507069	0.0572335	2.633194	0.0084586
income_group^8	0.1624476	0.0634318	2.560980	0.0104377
log_threeway_calls	-0.1320883	0.0516818	-2.555799	0.0105944
log_blocked_calls	0.0533137	0.0208965	2.551322	0.0107315
responds_to_mail_offers1	-0.3134609	0.1342850	-2.334296	0.0195802
prizm_codeOuter regional	-0.1154083	0.0496772	-2.323162	0.0201705
income_group^4	-0.1536455	0.0696182	-2.206973	0.0273159
age_unknown1	0.1553014	0.0712086	2.180935	0.0291882
occupationOther	-0.2417488	0.1138843	-2.122757	0.0337742
income_group.L	-0.1597670	0.0779188	-2.050430	0.0403225
log_inbound_calls	-0.0439096	0.0214323	-2.048757	0.0404859
income_group^5	0.1344574	0.0664682	2.022882	0.0430854
occupationProfessional	-0.2334525	0.1178357	-1.981170	0.0475722

Appendix 2.3: Performance Metrics for Logistic Regression Models (50% threshold)

	LR "All"	**LR "Log Selection"**
Accuracy	0.7178	0.7175
Recall/Sensitivity	0.0346	0.0330
Precision	0.0529	0.4970
F1	0.0648	0.0619
Kappa	0.0295	0.0277

Appendix 2.4: List of Predictors under Gini and Boruta methods

Method	Predictor Variables	
Gini	"log_current_equipment_days"	"log_equipment_age_relative_to_tenure"
	"log_revenue_per_minute"	"log_monthly_revenue"
	"log_monthly_minutes"	"log_call_issue_rate"
	"log_months_in_service"	"log_unanswered_calls"
	"log_outbound_calls"	"log_dropped_calls"
	"log_total_recurring_charge"	"log_overage_rate"
	"log_blocked_calls"	"log_inbound_calls"
Boruta	"handset_web_capable"	"buys_via_mail_order"
	"responds_to_mail_offers"	"has_credit_card"
	"income_group"	"credit_rating"
	"age_unknown"	"log_monthly_revenue"
	"log_monthly_minutes"	"log_total_recurring_charge"
	"log_overage_minutes"	"log_roaming_calls"
	"log_dropped_calls"	"log_blocked_calls"
	"log_unanswered_calls"	"log_customer_care_calls"
	"log_threeway_calls"	"log_outbound_calls"
	"log_inbound_calls"	"log_call_waiting_calls"
	"log_months_in_service"	"log_active_subs"
	"log_handsets"	"log_current_equipment_days"
	"log_age_hh1"	"log_retention_calls"
	"log_retention_offers_accepted"	"log_call_issue_rate"
	"log_revenue_per_minute"	"log_overage_rate"
	"sa_market_group"	"log_equipment_age_relative_to_tenure"

Appendix 2.5: Performance Metrics for Random Forest Models (50% threshold)

	RF "All"	RF "Log Selection"	RF "Gini"	RF "Boruta"
Accuracy	0.7189	0.7162	0.7104	0.7197
Recall/Sensitivity	0.0278	0.0266	0.0664	0.0402
Precision	0.5433	0.4552	0.4188	0.5495
F1	0.0528	0.0502	0.1146	0.0750
Kappa	0.0261	0.0197	0.0405	0.0379

Appendix 2.6: Performance Metrics for XGBoost Model

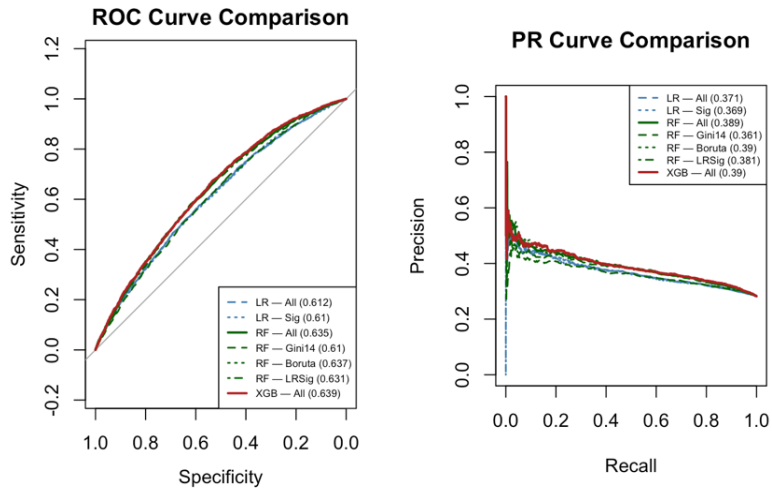
	XGBoost
Accuracy	0.7113
Recall/Sensitivity	0.1408
Precision	0.4630
F1	0.2160
Kappa	0.0970

Appendix 2.7: Selected Model (RF "Boruta") Confusion Matrix

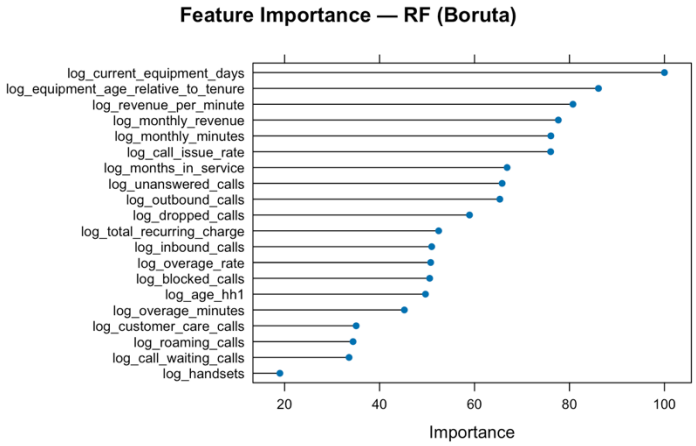
Confusion Matrix — Random Forest (Boruta) @ Threshold = 0.26

	Actual	
Predicted \ Actual	**No**	**Yes**
No	2632	562
Yes	3684	1923

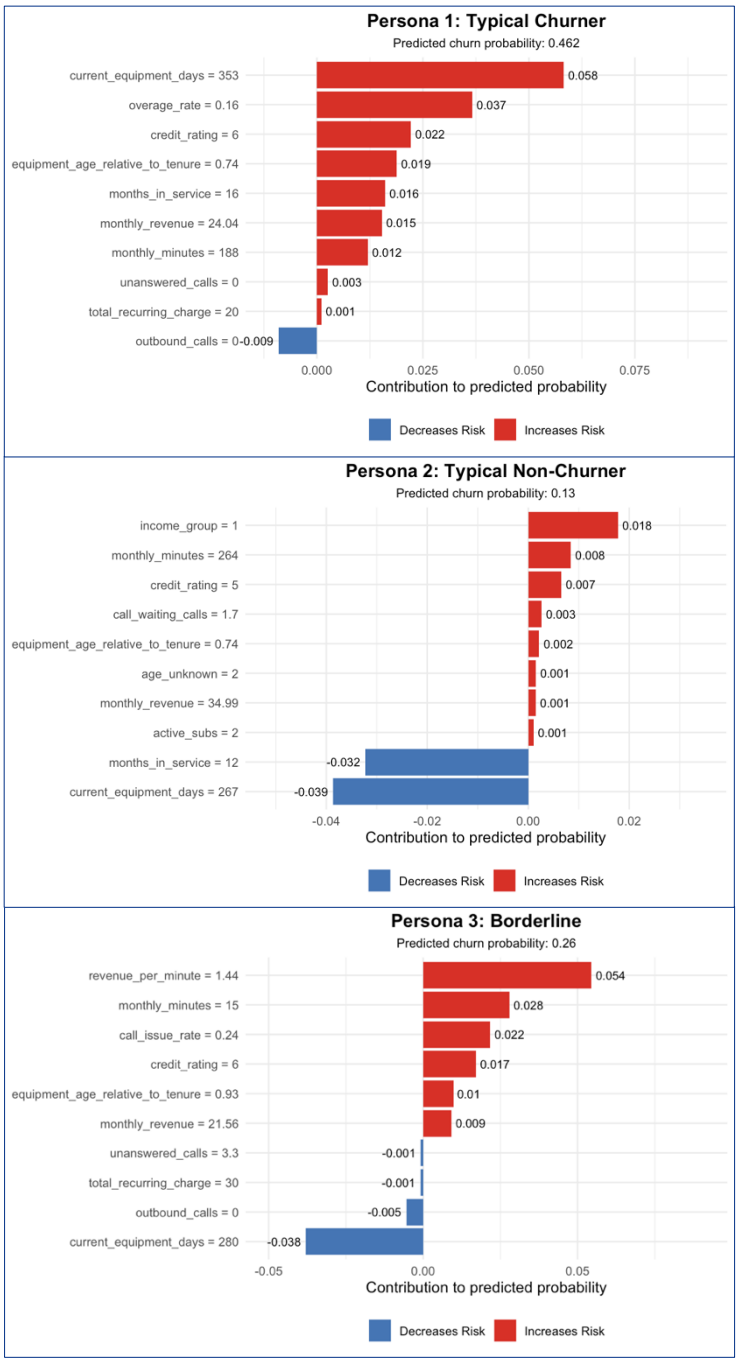
Appendix 2.8: Comparison of ROC Curves for Different Models



Appendix 2.9: RF "Boruta" Feature Importance Plot

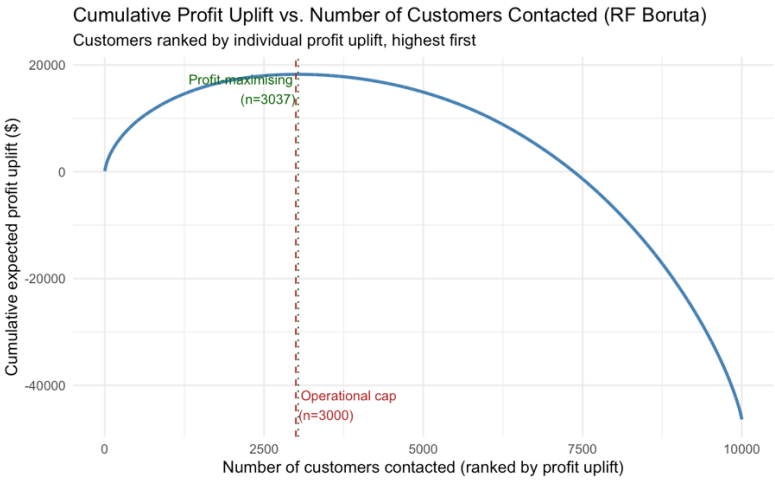


Appendix 2.10: Breakdown Plots for Customer Segmentation



Appendix 3 – Decision Optimisation

Cumulative Profit Uplift vs. Number of Customers Contacted Plot



Appendix 4 – Stochastic Modelling

Appendix 4.1: Convergence Table based on Simulation Size

No. of Simulation Runs	95th percentile total churns	5th percentile total profits
100	145.05	200415.2
250	143.55	200698.2
500	143.05	200805.4
2000	143.00	200695.8
5000	144.00	200783.6
10000	144.00	200790.3
20000	144.00	200760.6

Appendix 4.2: Total Profit and Total Churn Distributions

